

## Understanding adolescent gifted girls with ADHD: motivated and achieving

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The manifestation of ADHD in girls who are gifted can place strains on motivation and academic performance as they enter their middle school years. The purpose of this collective case study research was to examine the lived experiences of five girls who are gifted with ADHD in order to gain an understanding of the array of coping mechanisms used by this population as they navigate the academic pressures of their secondary school years. The girls in this study participated in a variety of educational settings including public, private, and charter schools. Using the experience sampling method, findings were drawn from the analysis of multiple data sources including experience sampling, document analysis, surveys, and interviews, in areas related to motivation as a result of their feelings toward school, relationships with teachers and family, and strategies used to center and refocus during times of stress.

**Keywords:** gifted/ADHD girls; twice exceptional; motivation

*Divergent:* People [who] differed from the standard and were able to use their brains and emotions in differing ways depending on the situation to decide for themselves rather than being stuck on one path. People did not become Divergent, they were born that way. ([www.divergentfactions.com](http://www.divergentfactions.com))

Veronica Roth's (2011) popular young-adult series describes a dystopian society in which youth are categorized and live their lives based upon their predetermined dominate virtue. Those who did not fit so neatly into a particular box were called divergent – forced to hide who they were from those around them. This same phenomenon occurs daily in schools all across the country. Scholars have estimated that as many as one-third of the students identified for special education services also exhibit gifted behaviors (Renzulli & Reis, 2009). Unfortunately, because they diverge from the norm, many of these gifted students go unrecognized, with their needs unmet by teachers, special education professionals, and administrators who are more often focused on students' weaknesses (Assouline, Nicpon, & Huber, 2006; Baum, Cooper, & Neu, 2001; Schultz, 2012). This results in school days spent with few, if any, educators recognizing these students' talents and in many cases even their challenges (Baum & Olenchak, 2002; McCoach, Kehle, Bray, & Siegle, 2001). This is known as the masking effect and it can become particularly relevant if the student with these characteristics also happens to be a gifted girl with Attention Deficit Hyperactivity Disorder (ADHD).

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It has been estimated that the clinical identification of male students as ADHD outpaces their female counterparts at a rate of 4:1 (Bauermeister et al., 2007; Cuffe, Moore, & McKeown, 2005). In 2013, the Centers for Disease Control reported that a total of 13.2% of boys have been diagnosed with ADHD, compared with only 5.6% of girls. The limited extant literature on girls with ADHD (e.g. Blachman & Hinshaw, 2002; Thurber, Heller, & Hinshaw, 2002) has reported the pressures faced by girls (e.g. friendship stability; self-esteem; academic achievement) in their secondary education years can be exacerbated by their ADHD and that these pressures are distinctly different from those faced by their male counterparts and can often lead to decreased motivation and academic achievement.

In the field of gifted education, there exists an inequity of gifted/ADHD research between the sexes with more attention focused on boys because they are more frequently identified with ADHD, while girls go unidentified, suffering in silence (Bauermeister et al., 2007; Cuffe et al., 2005). However, a review of the gifted literature and ADHD literature yields several similarities between girls who are gifted and those who have ADHD. Specifically, these two populations of girls face pressures that can effect motivation related to gender roles and expectations (Fleming, 2000; Hébert, Long, & Speirs Neumeister, 2001; Kerr & Vuyk, 2013); family, peer, and teacher relationships (Bianco, Harris, Garrison-Wade, & Leech, 2011; Grskovic & Zentall, 2010; Hébert et al., 2001; Mikami, Chi, & Hinshaw, 2004; Mikami & Hinshaw, 2006); and self-esteem and academic self-efficacy (Mikami & Hinshaw, 2006; Mikami et al., 2004; Zalecki & Hinshaw, 2004).

### **Influences on motivation**

Self-efficacy, self-concept, self-competency, and expectancy beliefs have been found to affect student motivation (Bandura, 1993; Eccles & Wigfield, 2002). Additionally, strong interpersonal relationships between teachers and students increases students' motivation and sense of belonging in school, helping them to take on more challenging academic endeavors (Gentry, Hu, Peters, & Rizza, 2005; Gentry, Steenbergen-Hu, & Choi, 2011; Patrick, Anderman, & Ryan, 2002; Powers, 2005). Finally, increased academic motivation has also been tied to parental involvement and support (Hebert & Reis, 1999; Schunk & Pajares, 2009).

### **Temperament and control**

Researchers and scholars have noted the positive and negative effects of temperament (e.g. Elliot & Thrash, 2002; Roskes, Elliot, & De Dreu, 2014) and control (e.g. Jewell & Kidwell, 2005; Patrick, Skinner, & Connell, 1993) on motivation. Rothbart, Ahadi, and Evans (2000) defined temperament as "individual differences in reactivity and self-regulation" (p. 123), stating that reactivity regulates behaviors (e.g. aggression and mood); whereas self-regulation provides control over attentional systems (e.g. effortful control and pleasure sensitivity). Further, researchers have noted that temperament can affect whether a student has an approach or avoidance motivational perspective (Elliot & Thrash, 2002). Approach motivation occurs when temperament is influenced by positive experiences that bring the student happiness or pleasure; whereas, avoidance motivation may be experienced when students are faced with undesirable events, causing them to take steps to escape participation in tasks that could potentially lead to failure (Elliot & Thrash, 2002; Roskes et al., 2014).

Roskes et al. (2014) discussed potential problems that arise when students are faced with situations that elicit avoidance motivational tendencies including anxiety, depression, and underachievement. The authors propose three possible strategies for the classroom that can improve the outlook and performance of students who are avoidance-motivated, giving them a greater sense of control over their academic success. The authors suggested that removing distractors would give students more control over their attentional behaviors, allowing them to focus on relevant classroom tasks. Additionally, they stated that avoidance-motivated students might benefit from environments that provide them with structure and focus, providing them with “specific directions and concrete ways to handle tasks” (p. 135). Finally, the authors discussed the importance of creating opportunities for these students to “replenish and reinvigorate” (p. 136) including opportunities to momentarily step away from stressful situations, providing feedback on progress, and encouraging positive mood states.

Patrick et al. (1993) conducted a series of studies examining the effects of perceived control and autonomy in the classroom on the motivation of 246 students, ages 8–10 years. Their results showed that strong positive motivation can occur when students not only feel a sense of control over their own learning, but also when they are provided with a sense of autonomy and choice in the classroom. Additionally, in their study of 170 college students, Jewell and Kidwell (2005) found that when perceived control was low, the gap between their perceived ability and their actual ability was significantly greater and resulted in lowered motivation than it was in students whose perceived control was high. Because students with ADHD frequently experience academic challenges (Zentall, 2006); gifted, adolescent girls have unique challenges (Bain & Bell, 2004; Kerr, Vuyk, & Rea, 2012); and gifted ADHD girls are largely absent from the literature (Fugate, 2014), exploring motivational influences of a sample of twice-exceptional girls is timely and important.

### *Gifted girls*

Frey (1998) found that the psychosocial needs and traits of gifted students can differ by gender, specifically in the areas of ownership of giftedness; dissonance between their own belief systems and the actions of others; risk taking; others’ expectations; impatience; identity; and sexuality. Reis (2002), in a chapter on gifted girls, reported several external and internal barriers from the literature that can affect motivation and achievement among gifted girls. Specifically, she pointed out that these girls valued their parents’ opinions, and any negative comments, real or perceived, then had a negative effect on their motivation and self-concept. Additionally, she found that teachers often believed and reinforced the idea that ability was innate in boys, but that girls must “work hard,” leading these girls to a loss of belief in their own abilities. She also discussed how gifted girls might avoid competitive situations in an attempt to preserve their relationships with their peers. These barriers may cause gifted girls to experience a lack of self-confidence and feelings of isolation (Hébert et al., 2001). Sadker and Sadker (1994) had this to say about the effects such barriers have on girls:

Each time a girl opens a book and reads a womanless history, she learns she is worthless. Each time the teacher passes over a girl to elicit the ideas and opinions of boys, that girl is conditioned to be silent and defer. As teachers use their expertise to question, praise, probe, clarify, and correct boys, they help these male students sharpen

ideas, refine their thinking, gain their voice, and achieve more. When female students are offered the leftovers of teacher time and attention, morsels of amorphous feedback, they achieve less. (p. 13)

Further, the transition from elementary to middle school can be particularly difficult for gifted girls as they begin to face pressure to better fit in with their peers (Bain & Bell, 2004; Dai, 2002; Kerr et al., 2012; Rimm, 2002), resulting in the potential for decreased motivation to achieve (Reis, 2002). It is around this time that many gifted girls begin to be faced with the choice of being smart or being popular, seeing the two choices as mutually exclusive (Kerr et al., 2012). However, in a qualitative study of seven gifted girls transitioning into sixth and seventh grades, Pepperell and Rubel (2009) found contradictory evidence that the transition may not be as difficult as the literature would suggest. These researchers found that the girls in their study had a strong sense of themselves and their abilities. The girls were able to find a balance between their giftedness and their social interactions through participation in extracurricular activities such as sports, theater, and student government. It was this balance that helped to keep them academically motivated while they found their place in the middle school social schema.

### *Perfectionism*

Perfectionist tendencies can be a risk factor for negative motivational and social-emotional development in gifted students (Tan & Chun, 2013). This can be particularly true for gifted girls who may set unreasonable or unattainable goals for themselves (Reis, 2002). However, Silverman (1999) argued that, "Perfectionism needs to be appreciated as a two-edged sword that has the potential for propelling an individual toward unparalleled greatness or plummeting one into despair" (p. 216).

Speirs Neumeister (2004) identified two types of perfectionism that affected how gifted students perceived their achievement and motivation – socially prescribed and self-oriented. *Socially prescribed perfectionists* are motivated by a fear-of-failure, often leading gifted students to develop performance-avoidance goals, with students avoiding challenging situations in which they felt they would fail. Alternately, this fear of failure might lead these students to develop performance-approach goals, engaging in rigorous and intensive coursework out of fear that if they did not, they would be perceived as inferior. These students often suffer from increased anxiety due to the perceived expectations of others (Reis, 2002). *Self-oriented perfectionists* are motivated to achieve, often seeking out challenges. It is this motivation to achieve that causes these students to develop mastery goals, focused on increasing skills and mastering tasks for self-improvement and understanding. Performance approach goals may also be developed, but out of a healthy sense of competition with peers. Students with a positive sense of perfectionism are more independent, resilient, supportive of like others, and involved in multiple activities (Reis, 2002).

### *Girls with ADHD*

The manifestation of ADHD characteristics in girls can result in lowered self-esteem, heightened emotional reactions, a lack of focus, and difficulty with peer and family relationships (Greene et al., 2001; Vail, 2002). These effects may be intensified as these girls enter adolescence, and can place additional stress on academic

motivation and personal performance (Owens, Hinshaw, Lee, & Lahey, 2009), leading to underachievement (Reid & McGuire, 1995; Reis & McCoach, 2000). Grskovic and Zentall (2010) studied 262 girls with and without ADHD. These researchers found that girls with ADHD were verbally impulsive, hyperactive, faster in conversation and school work, easily bored, impatient, and more easily prone to moodiness, anger, and stubbornness than their peers without ADHD. However, participants with ADHD in this study were also able to relate pro-social behaviors (e.g. participation in school activities and controlling impulsivity) with high self-esteem. Specifically, these girls were able to discriminate between their own appropriate and inappropriate behaviors, understanding “that they were more likely to react with strong feelings than other groups of girls” (p. 181).

Additionally, Mikami and Hinshaw (2006) and Owens et al. (2009) conducted studies of girls identified as ADHD who were 6–12 years of age. These researchers found that across time, long-term outcomes included increases in internalizing and externalizing symptoms, increased underachievement, higher rates of substance abuse and eating disorders, lower social skills, and difficulty maintaining peer relationships.

The socialization process becomes particularly challenging for girls with ADHD who have higher rates of peer rejection than their male counterparts (Blachman & Hinshaw, 2002; Carlson, Tamm, & Gaub, 1997) and “often serve as negative social catalysts, fueling conflictual social interactions among their peers” (Blachman & Hinshaw, 2002, p. 625). The exhibition of ADHD behaviors in girls has been found to be setting specific, with girls more likely to repress ADHD behaviors in academic settings where teacher disapproval is more likely. However, these behaviors become more prominent in social settings that “involve more complex rules and requirements, which may involve delayed and indirect consequences” the result of which is increased peer rejection (Grskovic & Zentall, 2010, p. 170). Mikami et al. (2004) found that, because of their social difficulties, ADHD girls run the risk of being “doubly disliked” by both peers and adults (p. 158). These girls have a heightened awareness of the social consequences of their behaviors, which can result in a poor perception of relationships with their teachers, lowered motivation and self-esteem, and increased rates of depression and anxiety (Rucklidge & Tannock, 2001).

Because ADHD girls are at greater risk of social impairment, when rejected by their peers they have increased possibilities of poor adolescent and adult adjustment that often lead to depressed and/or anxious behaviors (Gaub & Carlson, 1997; Greene et al., 2001; Mikami & Hinshaw, 2006). Researchers have identified links between peer rejection and academic underachievement, and increased incidents of substance abuse within this population (Barkley, Fischer, Smallish, & Fletcher, 2006). Further, if issues related to ADHD are left unaddressed, these girls find themselves at higher risk for low self-esteem (Becker, McBurnett, Hinshaw, & Pfiffner, 2013) and teen pregnancy (Quinn, 2005). These problems are likely to continue into adulthood where they have been found to experience higher divorce rates, financial problems, and difficulties with time management (Nadeau & Quinn, 2002; Rucklidge, Brown, Crawford, & Kaplan, 2007). However, Mikami and Hinshaw (2006) also found that girls with ADHD who were more confident in their academic abilities demonstrated lowered incidences of these negative results and actually demonstrated increases in achievement over time.

***Twice-exceptional learners***

Many twice-exceptional students are overlooked as a result of the masking effect because, though they may be working below their potential, their performance may not be below that of their grade-level peers (Hughes, 2011). This can have serious consequences on the academic, motivational, and social-emotional well-being of these students. Additionally, the characteristics of students with specific learning differences can be very distinct. Students with ADHD share characteristics similar to those associated with giftedness. For instance, Fugate, Zentall, and Gentry (2013) found that students who were gifted with characteristics of ADHD had higher levels of creativity – a trait often connected with giftedness – than their non-ADHD, gifted peers. Though scholars identify these similarities as a problem that they believe may lead to misdiagnosis (Baum & Olenchak, 2002; Cramond, 1995; Leroux & Levitt-Perlman, 2000; Moon, 2002; Webb et al., 2005), no empirical research exists regarding the frequency it may occur.

In 1988, Baum and Owen conducted a study with 112 gifted and/or learning disabled students. They discovered that although twice-exceptional students often had higher levels of creative potential, they had a tendency for disruptive behaviors and achieved at lower levels than their peers. These disruptive behaviors can be an outward sign of the student's feelings of inadequacy and low self-esteem. In their qualitative examination of four twice-exceptional students, Baum et al. (2001) found that these students often displayed emotional intensity and lacked self-confidence, likely because their weaknesses had been the focus in school. Other scholars have suggested that twice-exceptional students can experience extreme depression, have a sense of inferiority, and experience frustration leading to a general lack of motivation (Hughes, 2011; Moon & Reis, 2004; Olenchak & Reis, 2002).

Researchers have stated that relationships between students and teachers are a primary factor for increased student motivation and autonomy, resulting in decreased disruptive behaviors in the classroom (Anderman & Freeman, 2004; Davis, 2003; Patrick et al., 2002; Powers, 2005). Additionally, Townend, Pendergast, and Garvis (2014) reported that low motivation and poor relationships with teachers resulted in poor self-concept in twice-exceptional students. Unfortunately, teachers can sometimes associate behaviors commonly found among twice-exceptional learners, such as inconsistent performance and poor organization in school, with a lack of effort or laziness (Reis & Colbert, 2004). Beyond just being “nice,” teachers need to provide these students with a learning environment that will increase motivation, learning, and cognitive development and help them feel academically successful (Davis, 2003, p. 212).

Finally, Speirs Neumeister, Yssel, and Burney (2013) reported that parents of twice-exceptional students were among the first to recognize their child's intellectual abilities; among the first to recognize their disability; provided and advocated for support to meet their child's needs regardless of costs or convenience; sought to normalize the child's differences while maintaining high expectations for their performance; and helped their child understand the importance of self-advocacy and developing compensatory strategies to mitigate their differences. The benefits to students whose parents were actively involved in their education included increased motivation, better school attendance, higher grades, and fewer problems related to school transitions (Epstein & Van Voorhis, 2001).

Though limited in number, studies of gifted students with ADHD inclusive of both sexes exist. However, no work specifically addresses the unique motivational and achievement needs of girls with these exceptionalities. To address this gap in the extant literature, we examined how girls who are gifted with ADHD cope with anxieties that can affect their motivation and achievement as they navigate through their middle and high school years.

## **Method**

In collective case study, the researcher selects one area of concern, but selects multiple, representative cases for inclusion in the study and replicates the study procedures for every case. The integration of both the participants' point of view and the researcher's scientific perspective allows for the use of abductive reasoning to synthesize the meaning and understanding of individual cases and to identify similarities and differences between those cases (Stake, 1995).

### ***Participant selection***

When conducting collective case study, Creswell (2013) suggested that four to five participants are sufficient to allow for an in-depth exploration of their lived experiences. Through the process of purposeful maximal sampling (Creswell, 2013), five girls who were identified as gifted and who were diagnosed with ADHD were recruited from various secondary school settings around the USA, including public, charter, and private schools. Multiple different settings allowed consideration of the influence of teachers and parents on the motivational, emotional, and academic lives of these students. The rationale for this sampling procedure was to select representative cases to establish emergent themes and patterns within the various environments and to identify consistencies and exceptions within the findings (Creswell, 2013). The average age of all participants at the time of the study was 12.6 years with four of the girls identifying as Caucasian and one identifying as Hispanic. Three of the girls were taking medications for ADHD symptoms during their participation in the study.

Initial demographic information was provided on a participant profile sheet. Additionally, participants were asked to provide documentation that qualified professionals had identified them as gifted and as ADHD. Specifically, participants in this study provided proof of identification as gifted in their school or through their performance on standardized norm-referenced assessments (i.e. ability score of 120 or greater; overall 90th percentile or greater on normed tests of achievement or ability; or 70th percentile or above on any one sub-test). Proof of an ADHD diagnosis by a qualified physician or psychologist was also provided. Four of the girls had been identified as gifted by school personnel, with ability scores ranging from 121 to 141. Lea's ability score was 111 however; additional assessment qualified her for participation in this study by scores achieved in the subtest clusters of Phonemic Awareness (99th percentile, advanced), Verbal Ability (90th percentile, advanced), and Thinking Ability (70th percentile, average) on the Woodcock-Johnson III Cognitive Abilities Tests – Normative Update (Woodcock, McGraw, & Mather, 2007). These artifacts were used as part of the document analysis and provided an additional source for triangulation (Creswell, 2013). Table 1 provides a summary of the participant demographics including their chosen pseudonyms, ages, grade levels,

Table 1. Summary of participant demographics.

| Pseudonym        | Age | Ability score <sup>a</sup> | Grade | Race      | Type of school | Medication | State |
|------------------|-----|----------------------------|-------|-----------|----------------|------------|-------|
| Grace            | 12  | 132                        | 7     | Caucasian | Public         | Y          | IN    |
| Jenny            | 13  | 121                        | 8     | Caucasian | Private        | N          | CT    |
| Lea <sup>b</sup> | 13  | 111                        | 7     | Hispanic  | Public         | Y          | IN    |
| Lily             | 12  | 123                        | 7     | Caucasian | Charter        | Y          | CO    |
| Teresa           | 13  | 141                        | 8     | Caucasian | Public         | N          | OR    |

<sup>a</sup>Note: Ability score obtained from: Grace – Wechsler Abbreviated Scales for Intelligence; Jenny – Woodcock-Johnson III Tests of Cognitive Ability; Lea – Wechsler Preschool and Primary Scale of Intelligence, Third Edition; Lily – Naglieri Nonverbal Ability Test; Teresa – Wechsler Intelligence Scale for Children – Fourth Edition.

<sup>b</sup>Because Lea's ability score was below the 120 requirement, individual test scores on the Woodcock-Johnson III Cognitive Abilities Tests – Normative Update were examined. Though her overall GIA was in the 50th percentile, this score was reflective of her limited performance on tests of Cognitive Efficiency and Working Memory (4th and 8th percentiles respectively), both typical factors in students with ADHD. Lea qualified for participation in this study by her scores achieved in the clusters of Phonemic Awareness (99th percentile, advanced), Verbal Ability (90th percentile, advanced), and Thinking Ability (70th percentile, average).

ability scores, ethnicity, the type of school they were attending (i.e. public, charter and private), and whether or not they were taking ADHD medications at the time of the study.

### ***Research design and data collection instruments***

A portion of the anecdotal and quantitative data collection method was modeled on the Experience Sampling Method developed by Csikszentmihalyi and Larson (1987) and used in a study of 200 talented teenagers to obtain an accurate account of the momentary changes in the environment that affected participants' thoughts, activities, and mental states (Csikszentmihalyi, Rathunde, & Whalen, 1993). Qualitative data were drawn from a variety of sources including demographic information collected from the participant profile sheets and through individual interviews with the study participants. Additional qualitative data were collected using the Experience Sampling Form (ESF) that contained narrative responses detailing the activities that respondents were participating in, and their feelings about those activities at specific points in time over the course of the study, both in school and out of school. Further quantitative data were collected regarding participants' general dispositions and personal beliefs of their temperament and behaviors (Early Adolescent Temperament Questionnaire; Ellis & Rothbart, 1999, 2001), and the control they had over their own learning in the classroom (Students Perceptions of Control Questionnaire; Wellborn, Connell, & Skinner, 1989). All quantitative data were summarized using descriptive statistics to enhance understanding of the qualitative profiles of the respondents' beliefs about themselves and to help triangulate and analyze the emerging themes.

Because researchers have highlighted the connection between temperament and motivation (e.g. Elliot & Thrash, 2002; Roskes, Elliot, & De Dreu, 2014; Rothbart, Ahadi, & Evan, 2000), participants completed the Early Adolescent Temperament Questionnaire-Revised (EATQ-R; Ellis & Rothbart, 1999, 2001), a series of scales that measure temperament (activation control, affiliation, attention, fear, frustration, high intensity pleasure, inhibitory control, perceptual sensitivity, pleasure sensitivity, and shyness), and behavior (aggression and depressive mood). Ellis and Rothbart

(2001) reported sub-scale internal consistency estimates ranging from .65 to .82 for a sample of 176 adolescents – 82 boys and 94 girls – ages 10–16. Further, exploratory factor analysis of the scales resulted in factors of effortful control, surgency, affiliativeness, and negative affect, and girls scored significantly higher than boys on affiliativeness and depressive mood (Ellis & Rothbart, 2001). Definitions for each of the factors as they are measured on the EATQ-R are provided in Table 2.

Participants also completed the Students Perceptions of Control Questionnaire (SPOCQ; Wellborn et al., 1989), consisting of 10 subscales that measure beliefs in *strategies* – effort, attributes (ability), powerful others, luck, and unknown; the extent of the *control* they feel they have in using these strategies; and their *capacity* to access these strategies through effort, attributes (ability), powerful others, or luck with each subscale consisting of 6 items equally divided into success and failure (Skinner, 1995; Skinner et al., 2009; Wellborn et al., 1989). The strategy construct consists of five subscales with alpha reliability estimates ranging from .71 to .86. The control construct consists of one subscale with an internal consistency estimate of .74. Finally, the capacity construct is measured using four subscales with coefficient alphas ranging from .54 to .85 (Fulton & Turner, 2008).

### Procedures

Once qualified for participation, information collected from participant profile sheets was recorded together with demographic information (e.g. age, grade, ethnicity,

Table 2. Definitions of the factors measured on the EATQ-R.

| Factor                               | Definition                                                                                                                                             |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Temperament scales</i>            |                                                                                                                                                        |
| Activation control                   | The capacity to perform an action when there is a strong tendency to avoid it                                                                          |
| Affiliation                          | The desire for warmth and closeness with others, independent of shyness or extraversion                                                                |
| Activity level                       | Participation in activities requiring high levels of physical activity                                                                                 |
| Attention                            | The capacity to focus attention as well as to shift attention when desired                                                                             |
| Fear                                 | Unpleasant affect related to anticipation of distress                                                                                                  |
| Frustration                          | Negative affect related to interruption of ongoing tasks or goal blocking                                                                              |
| High intensity pleasure/<br>Surgency | The pleasure derived from activities involving high intensity or novelty                                                                               |
| Inhibitory control                   | The capacity to plan, and to suppress inappropriate responses                                                                                          |
| Pleasure sensitivity                 | Amount of pleasure related to activities or stimuli involving low intensity, rate, complexity, novelty, and incongruity                                |
| Perceptual sensitivity               | Detection or perceptual awareness of slight, low-intensity stimulation in the environment                                                              |
| Shyness                              | Behavioral inhibition to novelty and challenge, especially social                                                                                      |
| <i>Behavioral scales</i>             |                                                                                                                                                        |
| Aggression                           | Hostile and aggressive actions, including person- and object-directed physical violence, direct and indirect verbal aggression, and hostile reactivity |
| Depressive mood                      | Unpleasant affect and lowered mood, loss of enjoyment and interest in activities                                                                       |

Source: Ellis and Rothbart (1999).

family members, birth order, disability status, current grades in school subjects, hobbies, and interests) into a file for each participant. The girls were given a study packet that contained a welcome letter, the EATQ-R, SPOCQ, and 24 copies of the Experience Sampling Form. Participants completed the EATQ-R and SPOCQ individually, outside of the regular school day to avoid interfering with their academic schedules.

Over the course of three months, participants were prompted via text message twice weekly at varying times, once during the school day and once during non-school time, signaling them to complete the Experience Sampling Form (ESF; Csikszentmihalyi et al., 1993) resulting in 24 responses collected from each participant. Narrative responses to ESF questions established the setting, time of day, location, the activities with which and the people with whom the participants were engaged, and their feelings at that particular point in time (e.g. self-esteem, mood, and feelings regarding the activities). The ESF “embodies the convergence of the perceiver with the world, a transaction of external and internal dimensions” (Csikszentmihalyi et al., 1993, p. 51). Sample items from the ESF are included in Table 3.

Finally, based on the ESF results, additional qualitative data were collected through individual interviews conducted at the end of the study, each lasting approximately one hour. The interviews followed a semi-structured protocol with the purpose of gaining more detailed understanding of the girls’ perceptions of themselves; their strengths and challenges; their families; and their school environments. Some of the interview questions were derived from the literature relating to the girls’ feelings regarding their abilities and talents, school, and relationships with teachers and family (e.g. What are some limits or barriers that keep you from doing as well as you would like to? What do you perceive to be the expectations your parents have of you in your academic and future career trajectories?), while other questions inquired about the coping strategies, if any, that they had developed to help them deal with motivational and academic issues (e.g. What strategies do you use to help you deal with difficult situations in school related to academics?). Still other questions were derived from the emergent assertions, themes, codes, and exceptions found in the data they had provided during the study (e.g. Do you feel that [the activities that you participate in] have benefited you both as a gifted student and one with ADHD? Why or why not? Who do you turn to when you are feeling frustrated

Table 3. Sample items from the electronic sampling form.

| Construct                  | Questions: When you received the text, ...                         |
|----------------------------|--------------------------------------------------------------------|
| Self-esteem                | How self-conscious were you?                                       |
|                            | Did you feel good about yourself?                                  |
|                            | Were you in control of the situation?                              |
|                            | Were you living up to your own expectations?                       |
| Mood                       | What were you thinking about?                                      |
|                            | How well were you concentrating?                                   |
|                            | Describe your mood.                                                |
| Feeling regarding activity | Was this activity important to you?                                |
|                            | Do you wish you had been doing something else?                     |
|                            | Were you satisfied with how you were doing?                        |
|                            | How important was this activity in relation to your overall goals? |

and need encouragement?). At the conclusion of each interview session, we recorded our reflections, including field notes made during the interview regarding specific answers and observations. These served as a personal journal to address potential researcher bias. Each interview was transcribed verbatim, and then reviewed and compared to the audio recordings. Corrections were made to the transcripts during this comparison as needed.

### ***Data analyses***

Descriptive quantitative analyses examining means and standard deviations of the data collected from the EATQ-R and the SPOCQ were conducted to create descriptive profiles regarding their beliefs and general dispositions for each participant and for the group to examine how these beliefs and dispositions might affect their academic motivation. By comprehensively reviewing these data, clear descriptions of the trends and exceptions that emerged for each participant and for the group were derived. Analysis of qualitative data collected from the ESF and during individual interviews used the constant-comparative method of phenomenological analysis (Patton, 2002).

As interpreters of the data, we attempted to connect what was known about girls who are gifted and those who have ADHD, with the findings about motivation and achievement among girls who are *both* gifted and diagnosed with ADHD. To this end, individual participant responses to questions on the ESF and from the interview were read as a whole using a microanalysis process to formulate general impressions. Responses were then organized and grouped by individual questions for theoretical sampling analysis. This process of axial coding (Corbin & Strauss, 1990) allowed for further analysis and interpretation of the data through the clustering and re-examination of common themes with the goal of developing assertions related to the experiences of the study participants (Patton, 2002). The purpose of these assertions was to make connections between the theories related to motivation in adolescent girls who are gifted and girls with ADHD, to build upon, identify exceptions to, and extend findings in the literature.

### ***Trustworthiness of findings***

Creswell (2013) recommended several validation strategies including triangulation through the use of multiple data sources, a review of the procedures and/or data analysis by other researchers, and member checking. Triangulation was achieved through the use and analysis of a variety of data collection methods that included document analysis to ensure eligibility of the participants, the use of multiple self-report surveys, and interviews. To ensure fidelity and credibility in coding, qualitative data were independently analyzed and coded separately by us and by a colleague with expertise in qualitative analyses. We conferred about emerging themes and determined whether agreement existed between us concerning our findings, discussing discrepancies until we reached full agreement. This system of checking and rechecking of the emerging themes added to the conformability of the results.

To enhance the trustworthiness of the analysis further, member checking allowed participants to correct, alter, or extend issues that arose from the analysis, reinforcing the dependability of the results (Guba & Lincoln, 1989). We took participant feedback into consideration realizing that although the member check may not

necessarily be correct and the analysis incorrect, member imputations must be seriously considered in the reevaluation of the analysis (Guba & Lincoln, 1989). Researcher bias was controlled using a researcher's journal, which allowed for reflection to balance "what [we] know" with "what [we] observe" (Borg, 2002). Additionally, consultation with experts in the fields of giftedness and ADHD provided additional perceptivity and assisted in keeping researcher bias in check. Finally, we compared the findings found in these data with results reported in the extant literature on gifted girls and girls with ADHD to determine the transferability of these themes to other girls who are both gifted and diagnosed with ADHD.

### Quantitative results

Participants completed the EATQ-R (Ellis & Rothbart, 1999), which measures the general temperament and behavior of the participants on a five-point Likert-type response scale. Results are reported for each of the sub-scales with higher scores indicating a greater tendency for the participant to demonstrate each construct. Table 4 provides a complete report of the individual and collective case means and standard deviations for each of the sub-scales. Individual bolded results indicate scores that are one or more standard deviations above or below the mean results of the normative sample. All of the participants in this study reported higher than normal behavioral subscale scores for aggression and depressive mood. Additionally, as might be expected, when reported sub-scale scores for temperament were examined for the group it was found that all of the girls reported lower levels of attention. However, when inhibitory control was examined, Grace ( $M = 2.36$ ,  $SD = 1.21$ ) and Lily ( $M = 2.45$ ,  $SD = 1.29$ ) reported lower than normal scores; whereas, only Teresa ( $M = 4.00$ ,  $SD = 1.10$ ) reported a score that was higher than the norm. Additionally, only Jenny and Lea reported scores in the normal range for activation control, with Grace ( $M = 2.30$ ,  $SD = .71$ ), Lily ( $M = 2.30$ ,  $SD = 2.83$ ), and Teresa ( $M = 2.80$ ,  $SD = 2.83$ ), all reporting scores that were lower than normal. Further, Lea ( $M = 4.33$ ,  $SD = .71$ ), Lily ( $M = 4.33$ ,  $SD = .87$ ), and Teresa ( $M = 3.78$ ,  $SD = .97$ ) each reported higher than normal scores on the frustration subscale. Finally, when pleasure sensitivity was considered, defined as the amount of pleasure associated with activities that are low in intensity, complexity, and novelty, only Jenny ( $M = 4.00$ ,  $SD = 1.00$ ) scored higher than normal, with all of the other participants scoring in the normal range.

Scores from the SPOCQ (Wellborn et al., 1989) measured participants' self-reported perceptions of control using a four-point Likert-type response scale. Higher reported strategy belief sub-scale scores indicated a belief that these strategies were more effective; whereas, higher reported capacity belief scores indicated the girls' beliefs regarding their ability to use each construct. Additionally, scores were reported to indicate the level of positive or negative feelings the girls felt about each of the constructs and their sub-scales. Table 5 shows the individual and collective case study means and standard deviations for the construct of control, and the strategy and capacity belief constructs of effort and attributes. Scores that are one standard deviation or more below the mean are denoted in bold. For control beliefs, Jenny ( $M = 2.50$ ,  $SD = 1.64$ ), Lea ( $M = 2.33$ ,  $SD = 1.51$ ), and Teresa ( $M = 2.50$ ,  $SD = 1.64$ ) each reported scores lower than the norm. When strategy beliefs were examined, although all of the girls reported subscales scores within the norm for effort, only Teresa ( $M = 1.83$ ,  $SD = .75$ ) reported a lower than normal score in her

Table 4. Individual, collective case, and normative descriptive statistics for the EATQ-R.

| Construct                            | Grace       |             | Jenny       |             | Lea         |             | Lily        |             | Teresa      |             | Total |      | Normative sample (Girls) |     |
|--------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------|------|--------------------------|-----|
|                                      | M           | SD          | M           | SD          | M           | SD          | M           | SD          | M           | SD          | M     | SD   | M                        | SD  |
| Activation control <sup>a,c</sup>    | <b>2.30</b> | <b>.71</b>  | 3.30        | .71         | 3.80        | 1.41        | <b>2.30</b> | <b>2.83</b> | <b>2.80</b> | <b>2.83</b> | 2.90  | 1.20 | 3.57                     | .58 |
| Affiliation <sup>c</sup>             | 3.63        | 1.19        | 3.63        | 1.60        | 3.25        | 1.28        | 3.25        | 1.28        | 3.38        | 1.60        | 3.43  | .91  | 3.92                     | .52 |
| Aggression <sup>a,b</sup>            | <b>2.36</b> | <b>1.03</b> | <b>2.64</b> | <b>1.50</b> | <b>2.18</b> | <b>.87</b>  | <b>3.18</b> | <b>1.08</b> | <b>2.00</b> | <b>1.26</b> | 2.47  | 1.20 | 1.76                     | .60 |
| Activity level <sup>c</sup>          | <b>2.17</b> | <b>.75</b>  | 4.00        | 1.55        | 3.67        | 1.03        | 3.67        | 1.51        | <b>1.83</b> | <b>1.33</b> | 3.07  | 1.48 | 3.61                     | .69 |
| Attention <sup>c</sup>               | <b>2.71</b> | <b>1.11</b> | <b>2.57</b> | <b>.79</b>  | <b>2.29</b> | <b>.79</b>  | <b>2.00</b> | <b>1.15</b> | <b>2.29</b> | <b>1.25</b> | 2.37  | 1.06 | 3.35                     | .55 |
| Depressive mood <sup>a,b</sup>       | <b>3.33</b> | <b>1.21</b> | 3.00        | 1.10        | <b>3.33</b> | <b>.52</b>  | <b>3.67</b> | <b>1.51</b> | <b>3.33</b> | <b>1.51</b> | 3.33  | 1.15 | 2.49                     | .64 |
| Fear <sup>c</sup>                    | 2.50        | .84         | <b>4.17</b> | <b>1.33</b> | 2.67        | 1.63        | 2.33        | 1.03        | 3.17        | 1.17        | 2.97  | 1.33 | 2.80                     | .77 |
| Inhibitory control <sup>c</sup>      | <b>2.36</b> | <b>1.21</b> | 3.00        | 1.00        | 3.45        | 1.04        | <b>2.45</b> | <b>1.29</b> | <b>4.00</b> | <b>1.10</b> | 3.05  | 1.25 | 3.25                     | .49 |
| Frustration <sup>c</sup>             | 3.56        | 1.01        | 3.56        | .73         | <b>4.33</b> | <b>.71</b>  | <b>4.33</b> | <b>.87</b>  | <b>3.78</b> | <b>.97</b>  | 3.91  | .90  | 3.07                     | .63 |
| Pleasure sensitivity <sup>c</sup>    | 3.00        | 1.29        | <b>4.00</b> | <b>1.00</b> | 3.86        | 1.07        | 3.86        | 1.46        | 3.57        | 1.27        | 3.66  | 1.21 | 3.19                     | .77 |
| High intensity pleasure <sup>c</sup> | 3.00        | 1.41        | <b>3.73</b> | <b>1.01</b> | <b>3.91</b> | <b>1.04</b> | <b>3.45</b> | <b>1.37</b> | 2.73        | 1.56        | 3.36  | 1.32 | 2.84                     | .60 |
| Perceptual sensitivity <sup>c</sup>  | <b>3.83</b> | <b>.98</b>  | <b>3.83</b> | <b>1.60</b> | <b>4.50</b> | <b>.84</b>  | 3.50        | 1.64        | 3.17        | 1.47        | 3.77  | 1.33 | 3.14                     | .66 |
| Shyness <sup>c</sup>                 | <b>4.80</b> | <b>.45</b>  | 2.00        | 1.00        | 2.20        | 1.30        | <b>1.40</b> | <b>.89</b>  | 3.00        | 1.87        | 2.68  | 1.63 | 2.88                     | .75 |

Note: Bolded scores denote results that are one standard deviation or more above or below the mean.

<sup>a</sup>Denotes behavioral ratings.

<sup>b</sup>Behavioral scale means and standard deviations reported for girls ( $n = 77$ ) in a sample of  $n = 148$  (Ellis, 2002).

<sup>c</sup>Temperament scale means and standard deviations reported for girls ( $n = 527$ ) in a sample of  $n = 1055$  (Muris & Meesters, 2009).

Table 5. Individual, collective case, and normative sample descriptive statistics for effort on the SPOCQ.

| Construct               | Grace       |            | Jenny       |             | Lea         |             | Lily |      | Teresa      |             | Total |      | Normative sample <sup>a</sup> |     |
|-------------------------|-------------|------------|-------------|-------------|-------------|-------------|------|------|-------------|-------------|-------|------|-------------------------------|-----|
|                         | M           | SD         | M           | SD          | M           | SD          | M    | SD   | M           | SD          | M     | SD   | M                             | SD  |
| <i>Control beliefs</i>  |             |            |             |             |             |             |      |      |             |             |       |      |                               |     |
| Control                 | 3.00        | 1.26       | <b>2.50</b> | <b>1.64</b> | <b>2.33</b> | <b>1.51</b> | 3.50 | .84  | <b>2.50</b> | <b>1.64</b> | 2.77  | 1.46 | 3.51                          | .64 |
| Positive                | 4.00        | .00        | 4.00        | .00         | 3.67        | .58         | 3.33 | 1.15 | 4.00        | .00         | 3.80  | .56  |                               |     |
| Negative                | 2.00        | 1.00       | 1.00        | .00         | 1.00        | .00         | 3.67 | .58  | 1.00        | .00         | 1.73  | 1.16 |                               |     |
| <i>Strategy beliefs</i> |             |            |             |             |             |             |      |      |             |             |       |      |                               |     |
| Effort                  | 3.00        | .89        | 3.67        | .52         | 3.33        | .52         | 3.00 | .63  | 3.50        | .55         | 3.30  | .24  | 3.28                          | .56 |
| Positive                | 3.67        | .58        | 3.33        | .58         | 3.67        | .58         | 2.67 | .58  | 4.00        | .00         | 3.47  | .64  |                               |     |
| Negative                | 2.33        | .58        | 4.00        | .00         | 3.00        | .00         | 3.33 | .58  | 3.00        | .00         | 3.13  | .64  |                               |     |
| Attributes              | 2.50        | 1.05       | 2.17        | .98         | 3.33        | .52         | 2.67 | 1.03 | <b>1.83</b> | <b>.75</b>  | 1.40  | .19  | 2.63                          | .60 |
| Positive                | 2.67        | .58        | 3.00        | .00         | 3.67        | .58         | 2.33 | 1.15 | 2.33        | .58         | 1.53  | .83  |                               |     |
| Negative                | 2.33        | 1.53       | 1.33        | .58         | 3.00        | .00         | 3.00 | 1.00 | 1.33        | .58         | 1.27  | .59  |                               |     |
| <i>Capacity Beliefs</i> |             |            |             |             |             |             |      |      |             |             |       |      |                               |     |
| Effort                  | <b>2.33</b> | <b>.82</b> | <b>2.50</b> | <b>.55</b>  | 2.67        | .82         | 2.83 | .75  | 2.67        | 1.37        | 2.60  | .85  | 3.19                          | .55 |
| Positive                | 2.67        | .58        | 3.00        | .00         | 3.33        | .58         | 3.33 | .58  | 3.67        | .58         | 3.20  | .56  |                               |     |
| Negative                | 2.00        | 1.00       | 2.00        | .00         | 2.00        | .00         | 2.33 | .58  | 1.67        | 1.15        | 2.00  | .65  |                               |     |
| Attributes              | <b>2.00</b> | <b>.89</b> | 2.50        | 1.64        | <b>2.33</b> | <b>1.03</b> | 2.83 | .98  | 2.50        | 1.64        | 2.43  | 1.37 | 3.09                          | .60 |
| Positive                | 2.33        | 1.15       | 4.00        | .00         | 3.00        | .00         | 3.67 | .58  | 4.00        | .00         | 3.40  | .83  |                               |     |
| Negative                | 1.67        | .58        | 1.00        | .00         | 1.67        | 1.15        | 2.00 | .00  | 1.00        | .00         | 1.47  | .64  |                               |     |

Note: Bolded scores denote results that are one standard deviation below the mean.

<sup>a</sup>Means and standard deviations reported for a normative sample of students ( $n = 200$ ) ages 9–12 (Skinner, Wellborn, & Connell, 1990).

belief in attributes, defined by the authors of this instrument as ability, as an effective strategy. Finally, when capacity beliefs were considered, only Grace ( $M = 2.33$ ,  $SD = .82$ ) and Jenny ( $M = 2.50$ ,  $SD = .55$ ) reported lower than normal scores; while only Grace ( $M = 2.00$ ,  $SD = .89$ ) and Lea ( $M = 2.33$ ,  $SD = 1.03$ ) reported lower than normal attributes scores.

### Qualitative results

Through the process of axial coding (Corbin & Strauss, 1990), narrative responses from the ESF and from the individual interviews were grouped, allowing for the examination of themes and subthemes that emerged from these data. The girls reported several findings, italicized and discussed below, that related to their own academic motivation as well as the effects of their relationships with their teachers and parents.

#### Academic motivation

The girls reported *challenges in school* related to being both gifted and having ADHD that affected their overall academic motivation, including problems with distractibility and boredom with tasks such as homework.

I have a hard time paying attention if there is something distracting ... if someone's doing something wrong or chewing gum loudly, that bothers me. So like it's the slightest things, so I have a hard time paying attention. (Grace, personal communication, February 8, 2014)

There's a lot of equipment around [the science classroom] and I'm like, "Oh, pretty!" and there's a fish tank next to me and I am, "Look at the pretty fish." Then [my teacher's] like, "What's the function of psytoplasm?" and I'm like, "Ummm, to distribute proteins?" And he's like, "No" and then I feel like an idiot. (Lea, personal communication, February 10, 2014)

I cannot focus on my homework. I don't ... it's not that I don't like it, it's just that after a while I get bored and I don't want to do anymore after a certain point. It's just the same material. We've been doing it years before, and I know it ... (Jenny, personal communication, January 16, 2013)

[My teacher] gave out a lot of homework that I couldn't really finish and if I did, I forgot to turn it in so I got marked off a lot for that. Since I was GT, most of the kids got stuff faster than me, so I wouldn't really get the material. I wanted to cry. (Lily, personal communication, December 10, 2012)

I spend a lot of time doing homework or attempting to do homework. There's a lot of time spent *not* doing homework but not doing anything else either because I still have homework to do, if that makes sense. (Teresa, personal communication, November 25, 2012)

I forget my homework and when I don't have that, I don't really know that [material] as much. (Lea, personal communication, February 10, 2014)

Additionally, *internalized symptoms related to their dual exceptionality* can affect motivation both positively and negatively.

We have this college book, the cover says *283 Good Colleges* and I found William and Mary that I want go to so I'm starting to get my grades up for that ... (Grace, personal communication, February 8, 2014)

Sometimes I'm liable to give up, but I can't. I have to push myself to go forward. I have to care. You have to care to succeed, if you don't, you may succeed but you won't get far. I've learned that there are times when I didn't care, and I got bad grades. That's going to show on my records for college. I have to get good grades to go to college. I have to keep pushing forward. (Lea, personal communication, February 10, 2014)

In my English class, I didn't do very well. I got a 94 last quarter and then for math I was struggling. I just wasn't getting what he said at all. I ended up getting an 83. I'm taking advanced math classes, Algebra 2 and Trigonometry, so I guess that's still good, but I could have done better. (Teresa, personal communication, November 25, 2012)

Further, *physical and creative outlets* are compensatory strategies for self-regulation in these girls.

I don't know what I would do without running because when I have energy, I feel like I'm gonna spaz out. I don't know. It tires me out, but then you get this runner's high, which is just an awesome feeling. And then I get home, I have enough energy I still can, like, function and do my homework. (Jenny, personal communication, January 16, 2013)

The last time I made a marionette out of a stuffed pig. I don't know, my mom says that I'm creative but mostly it just helps me calm down. (Lily, personal communication, December 10, 2012)

Finally, these girls understood that *ability alone is an ineffective strategy* for academic success, stating that that effort and resiliency were also important factors.

I just think that if I really work hard, I might make something of myself. When I want to be smart, I can. This is when I want to be. I think I'm really smart, but I actually have to work at it. (Jenny, personal communication, January 16, 2013)

I try my best and give my best effort even if I'm not very good at something. (Grace personal communication, February 8, 2014)

## **Teachers**

The girls report the importance of receiving *academic* and *motivational support* from their teachers. Responses highlight the differences in their feelings these girls have for teachers who seem to misunderstand the challenges that they face and those teachers who strive to meet their educational needs.

I kind of am slower at taking notes and I think that the teacher – because she's really good at science – she doesn't really know what it's like to not be good at science. (Lea, personal communication, February 10, 2014)

This is going to seem like a pretty insignificant one, but when teachers make me read out loud. I'll look over the words and I'll get really red in the face and I want to crawl into a hole. I want to sound like I know what I'm talking about but it's new material. I don't know, it just makes me nervous. (Jenny, personal communication, January 16, 2013)

[My Science teacher] just makes us feel really bad if we didn't do very good on something. We just learned about cells and I can't get the difference between osmosis, like going into the cell, and diffusion which is also just another way of inserting something into the cell. So she makes me feel really bad about that when I don't get it. It makes me feel discouraged with myself. It just makes me not like the class as much. (Lea, personal communication, February 10, 2014)

He goes over the material in ways that are interactive to the class. (Jenny, personal communication, January 16, 2013)

We're doing this book thing where we have to read a book that [our teacher] approves. They have to be award winning or notable. I asked [her] if Stephen King counts and she said yes. I was like, "YES!" I just adore him, his books, his writing; he's a great author. (Lea, personal communication, February 10, 2014)

She sort of gets that I need more time on things or [writing] notes for things. If she's going through something on the board, she'll give [a copy] to me, but not in a really obnoxious, obvious manner, which I appreciate. (Teresa, personal communication, November 25, 2012)

She encourages us to do our best, to work hard. Especially since some of the problems don't take a short time to do, they take some time to think about them. It makes me think about how I'm doing and if I understand what I'm learning ... She encourages me to check over my stuff and make sure that I have it as good as I can make it. (Grace, personal communication, February 8, 2014)

It's kind of a creative school. It's not all about math or science or reading or writing, it's about art too. [Since moving to this school] my grades have gone up so now I feel like I am smart and I can do more ... I think more of the kids at this school have ADHD too and the teachers understand that. [My English teacher] is so supportive. We have creative stories sometimes, and she says that I write really well. (Lily, personal communication, December 10, 2012)

### Parents

Finally, like teachers, the *academic* and *motivational support* that these girls received from their parents profoundly affected the academic achievement that these girls experienced, even when they felt increased pressure as a result of their perceptions of their parents' expectations.

Academically, if like, they hadn't helped me to have gotten my accommodations that have been very helpful – like double time especially – I would have done much worse. (Teresa, personal communication, November 25, 2012)

My dad is really the one that encourages me academically. We do a weekly grade check and stuff like that. (Grace, personal communication, February 8, 2014)

[My mom] expects me to get better grades and straight A's and I'm just like I can't do that and, I don't know, it's just – I get pushed so much and they've got really high expectations. They want me to have a good future, but I'm tired of being pushed all the time because I know I can do well, I expect myself to do well, too, it's just sometimes I want them to lay off and let me take care of it myself. (Jenny, personal communication, January 16, 2013)

My mom. I wouldn't admit it, but yes. She makes me work my butt off and, I mean, half the time that just annoys the crap out of me but I know she's doing it because she loves me. She's like, "Come on work harder!"; "Seriously, you got a C on that test, you did not just do that!" And then I have to work even harder next time because I don't want the wrath of her coming down on me, but if I didn't have her pushing me, I would not work as hard. (Jenny, personal communication, January 16, 2013)

### Discussion

The girls in this study reported several positive and negative influences on their motivation and academic achievement as a result of being both gifted and having

ADHD. These girls can sometimes associate school with feelings of confusion, tension, and shame, resulting in lower self-esteem and motivation. Nevertheless, they also described ways they deal with their challenges; accommodations from their teachers; and support from their families, all which motivate them to achieve. Distractibility and boredom hindered motivation and achievement however, this was mitigated when they felt that their teachers understood and supported their needs, a finding that underscores the importance of teachers who connect with their students (Gentry et al., 2011). Finally, although relationships with parents often provided positive motivation, the girls in this study also reported frustration when they felt that their parents' expectations were too high, supporting previous research that gifted girl's perceived perceptions of their parents' beliefs can affect their motivation (Reis, 2002).

### ***Academic motivation***

*Challenges in school*, including distractibility and failure to complete tasks such as homework were reported. Consistent with the literature on students with ADHD and students who are gifted with ADHD (Baum & Olenchak, 2002; Grskovic & Zentall, 2010; Moon, 2002; Zentall, 2006; Zentall et al., 2001) and self-reported higher than average perceptual sensitivity on the EAQT-R (Ellis & Rothbart, 1999; 2001), these girls have a difficult time maintaining focus and attention. Even the slightest distraction, such as the presence of a fish tank close by in the classroom, can divert the attention of the girl who is gifted with ADHD (Lea, personal communication, February 10, 2014).

Further, they become easily bored and unmotivated when faced with completing homework that they see as repetitive. Suddenly, time doing homework becomes time not doing homework; a fact that they are cognizant of, but they cannot bring themselves to do something else, so they end up spending time doing nothing at all. This is unsurprising given the scale-scores of responses to questions of activation control on the EATQ-R (Ellis & Rothbart, 2001) demonstrating a diminished capacity for gifted girls with ADHD to complete activities that they would rather avoid. When homework is completed, the chances that it makes it to the teacher are small, even when organizational systems are put into place. These girls are also keenly aware of consequences related of their lack of focus on their achievement, and this then has an adverse effect on their motivation.

*These gifted girls with ADHD also face many internalized symptoms* as a result of their co-occurring conditions. Even at the middle school level, many of these girls are already looking beyond high school and considering their options for college. Consistent with Baker's (Baker, 1987) findings that perfectionist tendencies are more prevalent in gifted girls than their non-gifted peers and because of the traits associated with their co-occurring ADHD, these twice-exceptional girls worry about their grades and become hypercritical of their own performance; a 94 in an English class and an 83 in an advanced math class are considered only okay with the girl wishing she had done better (Teresa, personal communication, November 25, 2012).

Fortunately, these girls are also often self-aware and able to reflect upon past failures and use them as positive motivation, traits that Neumeister (2004) identified in self-oriented perfectionists. Although this drive and determination positively motivated some of these twice-exceptional girls in some circumstances, unreasonably high expectations also resulted in low self-esteem and motivation in other

circumstances, supporting the findings of past researchers (Hébert et al., 2001; Neumeister, 2004; Reis, 2002). Thus, their motivation and self-image is tied to the challenges of ADHD.

*Fortunately, for these girls, physical and creative activities become important compensatory strategies for self-regulation.* To cope with the pressures of middle school, these girls turn to physical activities that help them refocus, calm, and center themselves. In addition to physical activity, having a creative outlet also helped to counteract issues related to distractibility, boredom, and low motivation. This supports the recommendation of Roskes et al. (2014) that students who are susceptible to avoidance motivation should be provided with opportunities to “replenish and reinvigorate” (p. 136), and extends this as a possibly important factor for the academic success and motivation for these gifted girls with ADHD.

Finally, as demonstrated by attributes scores on the SPOCQ (Wellborn et al., 1989), these gifted girls with ADHD believe that *ability alone is an ineffective strategy*. They place a similar value on hard work as a strategy as those included in the normative sample. Interestingly though, on this same scale, they averaged scores that were 1 SD below the mean when asked about their belief in their *capacity* to work hard, a fact that also came through in their interview responses. These gifted girls are aware of the challenges that they face as a result of their ADHD and are motivated to overcoming them. Their belief in their ability to persevere results in these girls having a growth mindset (Dweck, 2007), seeing their abilities as malleable and connected to their effort. This is in contrast to the findings of Dweck and others that girls tended to have more fixed mindsets focused on ability (Dweck, 2007; Dweck & Bush, 1976; Dweck et al., 1978; Good, Rattan and Dweck, 2012).

### **Teachers**

Twice-exceptional students face many obstacles in the classroom as a result of their learning, motivational, and/or behavioral differences (Baum et al., 2001; Hughes, 2011; Moon & Reis, 2004; Olenchak & Reis, 2002) that require *academic support* to address their giftedness and their ADHD. All too often, these gifted girls with ADHD find themselves in classroom settings where they feel alienated or misunderstood because teachers fail to fully realize the challenges that they face on a daily basis. Grskovic and Zentall (2010) discussed the academic difficulties that ADHD girls have, particularly in areas of math and science. Unfortunately for these gifted girls with ADHD, these can be some of their most unforgiving classes. Additionally, these girls are often asked to perform tasks that highlight their challenges, such as reading aloud in class. Consequently, when teachers lose patience with these girls or fail to recognize their giftedness, learning preferences, or needs; motivation and achievement decline.

Though incidents like these are surely unintentional, they highlight the need that exists for research-based professional development focused on differentiation and the needs of twice-exceptional learners in general, and gifted girls with ADHD in particular. The idea of differentiation to meet the needs of gifted learners in the classroom is not new (for review see Olenchak, 2001; Tomlinson, 2004). However, for these girls who are gifted with ADHD, differentiated instruction can provide the *motivational support* they need to increase their achievement, self-esteem, and self-efficacy. These girls require more hands-on and visual learning experiences that put discrete skills into a larger context that they can relate to and understand. In addition

to contextual, hands-on learning experiences, providing these girls with more opportunities to have choices in their learning honors their interests and increases their motivation (Patrick et al., 2014). These findings support the literature on twice-exceptional learning differences suggesting that these students are motivated to achieve when learning is presented from a strength-based perspective (Baum & Olenchak, 2002; McCoach et al., 2001; Neihart, 2000).

Just as important as personal accommodations made by their teachers are the relationships that educators develop with their students and the environment that they create in their classrooms and schools. First, when these girls feel that their learning differences are honored and supported, their motivation increases. Second, when these girls feel that their teacher genuinely cares, they want to work harder for that teacher and the byproduct is that they reflect upon their own learning. Finally, these girls need an environment in their classroom and in their school that meets their unique needs and encourages them to achieve.

### ***Parents***

However, these types of educational experiences are not always readily available to these girls. *This is where the academic support received from their parents becomes so important.* Parents have long been regarded as an important component in the motivation and academic success of gifted girls (Comallie-Caplan, 2008; Hébert et al., 2001; Kerr et al., 2012; Reis, 1998, 2002). These girls look to their parents to be their advocates to ensure that they are provided every opportunity for success. They appreciate the lengths that their parents go to make sure that they receive the accommodations that they need and that help them better understand themselves as gifted learners with ADHD.

In addition to their advocacy, parents provide valuable *motivational support*. Unfortunately, whether real or perceived, at times these twice-exceptional girls feel additional pressure as a result of what they see as unreasonable expectations from their parents; when this happens, these girls become highly frustrated wanting their parents to “lay off and let me take care of it myself” (Jenny, personal communication, January 16, 2013). This opens the door to the possibility of these girls developing socially prescribed perfectionist tendencies (Neumeister, 2004). Fortunately, these girls also seem to understand that even though they may become easily frustrated by their parents’ expectations, “... I know she’s doing it because she loves me” (Jenny, personal communication, January 16, 2013).

### ***In their own words***

Collective case study requires that researchers infer the meaning of the stories of our participants. To do this, we must collect and analyze data from a variety of sources understanding that, even controlling for our own personal biases, our interpretations are exactly that – interpretations. Therefore, we felt it important that our final interview question asked each of the girls what they wanted us to understand about their experience as a girl who is gifted with ADHD. They responded, without prompt, with thoughts concerning motivation, talking about hard work, success, and how they have learned to persevere. We report these final thoughts without analysis as they speak for themselves and indicate the transferability of the findings to other girls like them.

I want you to take away that people like me; we sort of have to work for it a little bit. With ADHD it makes you ... it makes you not focus as well on things, so we have to work harder for it. But if you work harder, you do better. Everyone is different. I don't think everyone who has ADHD and is gifted is like me; I don't think everyone that is gifted is like me. There's different ways for people with ADHD to be gifted. That's what I think. (Grace, personal communication, February 8, 2014)

It's hard. Social life is not going to be easy. Finding something that is going to be a good outlet for you makes you work hard. For me it's running. (Jenny, personal communication, January 16, 2013)

Once you find the right school or you get the right help, you can do a lot more. (Lily, personal communication, January 25, 2013)

Sometimes I'm liable to give up, but I can't, I have to push myself to go forward. If I don't – like some people might – but I have to care. You have to care to succeed, if you don't, you may succeed but you won't get far. I've learned that there are times when I didn't care and I got bad grades. That's going to show up on my college, um, things. So ... I have to get good grades to go to college. I have to keep pushing forward. (Lea, personal communication, February 10, 2014)

I'd say, like, just to start off, it's that people don't really understand because, like, I'm gifted a lot of people might not notice that I also, like, have trouble with something. And they think I'm stupid instead. Or they might know that I have ADHD or dyslexia, but they forget it and just get annoyed with everyday occurrences like me being slower on something or me fidgeting a little bit or people having to repeat things. I'll be listening and trying really hard, but then I might start to drift off and think of other things, which is why I take so much longer on other things. I think there's a lot of short-term memory things like with directions and little things that I'd like to remember that I suddenly forget. Or, like when I'm talking, I'm like, I was going to say that but I forget. I'm constantly thinking of other things, and then [there are] conversations with people where I might randomly say something else because I'm thinking of a bunch of things, or ... just in terms of like, um ... I'll like ... I'll be sort of lost, like ... um ... you know there's something that I'm not interested in, but other people will keep talking and then they might ask me a question or something like that and I have no idea what they're talking about. But I guess, just in general, it makes things harder. It makes everything harder to get through. And you have to be really determined to get through something and if not, you might not get through it. So I think, that's sort of like, it in general. (Teresa, personal communication, November 25, 2012)

## **Implications**

The findings from this study have implications for gifted girls with ADHD, their teachers, and their parents. First and foremost, these girls should understand that they are not alone. Girls who are gifted with ADHD have an acute awareness of their own strengths and weaknesses. Girls who are gifted with ADHD can learn from these girls that their dual exceptionalities make them unique and that they can leverage their energy and learn to adjust their motivation and learning through physical exercise and creative endeavors. It is important that these girls understand that although it is normal to feel a certain amount of fatigue from the frustrations that come from being gifted with ADHD, having an outlet, either physical or creative, can help them face their challenges and succeed, increasing their ability to achieve academically.

Educators should find ways to support the motivation and achievement needs of these gifted girls with ADHD. These girls have perfectionist tendencies that drive them to achieve, despite the challenges that they face as a result of their ADHD.

Educators need to recognize this desire to succeed but also understand that this is not always a result of positive motivation. It is important that they connect with and work with these girls and help them set reasonable and attainable goals while maintaining challenge and rigor in the classroom.

Academic support provided by caring teachers who recognize these girls' strengths and then use those strengths to address any academic challenges increases motivation and achievement. Introducing research-based approaches such as curriculum compacting (Reis et al., 1992) to classroom practices will capitalize on the girls' interests by allowing them to pursue self-directed avenues of learning while reducing the amount of redundant homework that occurs in subjects in which the girls demonstrate mastery and thereby decreasing chances for low self-esteem due to missed homework assignments. This, coupled with opportunities for choice in classroom assignments and projects, will help to positively motivate these girls to achieve academic success while decreasing the negative effects of tedious, uninteresting activities associated with repetitive homework. Finally, teachers should create classroom environments that are focused on developing relationships with and among their students. This trust can be an important factor in the academic and emotional success of gifted girls with ADHD. When these girls feel that their teachers not only understand their needs, but also honor them, their motivation, self-esteem, and self-efficacy increase.

Parents play a significant role in the motivational and academic success of their gifted daughters with ADHD. Girls who are gifted with ADHD can have co-occurring learning differences, such as processing speed or dyslexia. This has implications for parents who recognize these challenges and must find a balance between supporting their daughter's learning difficulties and maintaining high expectations for their academic success. Parents need to be aware that support can turn into pressure and too much pressure can undermine motivation and achievement. With adolescent twice-exceptional girls there is a fine line between perceived support and too much pressure. Finally, it is important that parents recognize the role of physical activity or creative outlets as mediators during times when these girls feel stressed. Encouraging them to take up a sport, go for a run, engage in arts, or find another physical and/or creative outlet can help gifted girls with ADHD begin to regulate their stress, which in turn may contribute to increased motivation to achieve in school.

### **Limitations**

In all qualitative research, there is concern regarding the validity of the data collected. Unlike Csikszentmihalyi et al.'s (1993) original study, there was no requirement that the girls respond to the ESF within a certain time period after receiving the text notification. This decision was made to limit disruption to the participants during school times when they may have been involved in academic activities. However, the overall quality and variety of data collection methods used in this study sought to yield trustworthy results (Creswell, 2013), framing the lived experiences of an often-ignored population of twice-exceptional girls during a vulnerable time of their lives. Finally, as a byproduct of qualitative research as well as the limited diversity of the girls, there may be limited generalizability of the results beyond that of those participating in the study. However, the use of collective case study methods involving girls from varying educational settings allowed for the emergence of

themes that unite these girls despite any differences that might have existed in their personal lives and as such, allows for greater transferability of the findings to other girls who are both gifted and ADHD.

## Conclusion

This study serves as a baseline for future inquiry. These girls who are gifted with ADHD have a unique ability to view the world through multiple lenses, rejecting the predetermined paths often associated with girls who are gifted and girls with ADHD. As such, we propose that researchers, educators, and parents stop seeing these girls as gifted with ADHD but rather as girls who are ADHG (Attention *Divergent* Hyperactive *Gifted*). Such a paradigm shift would then alter the focus from their challenges as girls who are gifted and as girls with ADHD and instead, highlight their motivation, strengths, perseverance, and resilience; those innate qualities that make them so very special. It is our hope that more researchers in the field of giftedness will take up this mantle and help remove the stigma that has kept these girls with ADHG hidden for far too long.

## Disclosure statement

No potential conflict of interest was reported by the authors.

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